Anticipating Demand, Enhancing Response

A Forecasting-Based Approach for Snow Removal System in Pittsburgh

Github repo: https://github.com/moidulhn/pittsburgh-snow-removal-forecasting

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1.Executive Summary

The city of Pittsburgh experiences moderate levels of snowfall, averaging around 44 inches every year. The actual snowfall every year fluctuates significantly, especially in recent years due to a multitude of different factors including climate change. This creates challenges in planning and response. Although Pittsburgh has a structured snow removal system in place, it is often reactive and can become strained during periods of heavy or unexpected snowfall. Forecasting the removal requests by area based on historical data and weather patterns could enhance operational preparedness by optimizing the deployment of snow removal vehicles (such as plows and salt trucks). This ensures improved response efficiency, equity, effectiveness and improved public safety overall.

Our project aims to address the current challenges faced by Pittsburgh's Department of Public Works (DPW) reactive snow removal system, which currently depends on 311 requests raised by residents. This approach fails during major storms as the response is often delayed with backlogs and uneven service across neighbourhoods- particularly affecting marginalized or geographically challenging areas. We propose the utilization of a machine learning based forecasting approach that predicts neighborhood-level snow removal requests in advance, enabling the city to shift from reactive to proactive service delivery.

The final model we implemented was a Gradient Boosting algorithm trained on historical data of 311 snow removal request data, weather conditions and traffic patterns. We managed to predict the daily complaint counts for each neighbourhood within a reasonable margin of error. The model was evaluated MAE, RMSE and MAPE error metrics. The model is especially effective during peak snow periods and in high-demand neighborhoods. The results are interpretable through SHAP feature importance analysis.

We recommend the predictive model be used by DPW to preemptively deploy crew members and salt distribution vehicles. Since our model generates daily, neighborhood-level forecasts, it provides the granularity needed for both short-term operational decisions and long-term planning. The model should be integrated into DPW's planning workflows, starting with a pilot deployment in select districts, and eventually expanding to real-time use across the city.

2. Background and Introduction

Pittsburgh's winter climate is marked by frequent snowfall, sudden temperature shifts, and freezing rain events, particularly from November through March. Due to its geologic location, the city experiences microclimates where steep hills and narrow streets lead to highly variable snow and ice accumulation. The City of Pittsburgh's Department of Public Works (DPW) follows a tiered snow removal protocol: treating primary routes first, then secondary roads, and lastly residential streets. Residents request snow removal services through the city's 311 system. However, this reactive approach often results in request backlogs, delayed response times, and uneven service provision, particularly during major storms, hard-to-reach or underserved neighborhoods, increasing public frustration and complaints.

This project aims to reduce such issues by enabling a more proactive, transparent, and equitable snow response across the city with a forecasting-based solution. It aims to predict 311 snow removal request surges before they happen, enabling proactive deployment of plow teams, better scheduling of staff, and early management of salt inventories. By anticipating demand geographically and temporally, the city can prevent backlogs, respond more equitably across neighborhoods, and communicate estimated response times to residents. The key actors who will benefit and potentially act on this work include the DPW, who manage plowing operations; city policy staff, who oversee equity and service standards; and the Mayor's Office, which is accountable to the public during service disruptions.

This project aligns with three central policy goals:

• Efficiency: By forecasting service demand, the city can better match staff and trucks to high-need areas, reducing overtime costs and idle deployment.

• Equity: Historical data will be used to identify service gaps, ensuring that marginalized or topographically difficult neighborhoods are not left behind during storms.

• Effectiveness: The forecasting model allows for measurable improvement in response times, issue resolution rates, and resident satisfaction, especially during peak storm periods. This creates a feedback loop where operational outcomes continuously improve based on model predictions and post-storm data audits.

3. Related work

Snow response in many U.S. cities, including Pittsburgh, is traditionally guided by a mix of historical knowledge, real-time weather forecasts, and manually reported 311 service complaints. These decisions are largely reactive and based on general operational guidelines, staff experience, and simple prioritization heuristics. Although tools such as plow tracking systems and salt distribution monitors are in place, they primarily serve observational or coordination purposes. There is currently no formal system for predicting the volume or spatial distribution of snow-related requests in advance.

Previous efforts to improve municipal snow management have primarily centered on predefined policy rules, such as ensuring main roads are cleared first or dispatching crews when snow accumulation exceeds a certain threshold. These approaches, however, do not account for temporal demand patterns or neighborhood-level variability in complaints. As a result, they often fall short in proactively allocating resources based on actual service needs.

Machine learning has been applied to various urban challenges such as crime forecasting, energy usage prediction, and public transportation demand modeling. However, within the context of 311 systems, most studies have focused on categorizing complaint types or optimizing response times, rather than forecasting complaint volumes across time and space.

This project introduces a novel approach by integrating over nine years of 311 snow complaint data with historical weather and traffic information. The resulting panel dataset operates at the daily and neighborhood level, allowing for predictive modeling of complaint demand before it occurs. Unlike prior methods, this framework enables anticipatory decision-making, with the potential to support more efficient and equitable snow removal strategies across the city.

4. Problem formulation and Overview of solution

The formulation of our problem is a supervised regression, where the target variable is the number of complaints on a given day and in a given neighborhood. The objective is to minimize forecast error across all locations and snow seasons, allowing the city to proactively plan staffing, equipment deployment, and routing logistics.

We model the complaint count $y_{n,t}$ for neighborhood n on day t as a function of several predictors:

• **Temporal features:** month, weekday, snow season indicator
• **Lagged behavior:** complaint counts from recent days (e.g., lag-1, lag-3), and rolling statistics (e.g., 3-day mean, 3-day std)
• **Weather variables:** daily snowfall, temperature extremes, precipitation
• **Neighborhood characteristics:** traffic volume, area size (optional for extended modeling)

We treat each neighborhood-day as an independent observation (panel structure), with feature vectors constructed from both same-day and past-day information. This enables the model to learn temporal patterns, spatial variation, and non-linear effects (e.g., threshold behavior under heavy snow). Our pipeline consists of:

1. **Data cleaning** of 311 complaints and weather records (2015–2024)
2. **Feature engineering** of temporal, weather, and autoregressive inputs
3. **Baseline construction** (Lag-1, Lag-7, Year-over-Year)
4. **Supervised learning model** (LightGBM)
5. **Evaluation** using MAE, RMSE, and MAPE on holdout years (2023–2025)
6. **Interpretability analysis** via SHAP and feature importance
7.

The result is a citywide and neighborhood-level forecasting system that provides more accurate, timely, and interpretable predictions than traditional heuristics.

5. Data Description

The primary dataset available for analysis is Pittsburgh's historical 311 snow removal request records. These include request attributes such as request type, creation and closure dates, request status (open or closed), department handling the request, geographic data including latitude, longitude, street, cross street, neighborhood, census tract, and council district. In particular, we collected all the data available and filtered for requests that were categorized under either **"Snow/Ice removal"** or **"Sidewalk, Lack of Snow/Ice Removal"**. The following were the observations we made about the data collected. We also used Python to analyze the dataset:

- The data contains records between April 2015 and December 2024 (~9 years) encompassing ~9 winters. There are **644 unique days** on which requests were made. This gives us enough data over the years to allow for a comprehensive analysis.

- We conducted descriptive analysis and found a total of 31,519 relevant requests. Among them, **30,228 (95.9%)** have a “closed” status. Given the number of records, we can safely assume that time series forecasting models (such as ARIMA or FB Prophet) trained on this data will generalize well.
- In terms of request frequency, the data has **a mean frequency of 49 requests per day** (for days on which reports were made). But this is most likely a skewed number as the median request frequency is 5 requests per day and the highest frequency is 1471 requests on December 17, 2020 made during the second snowiest December on record for Pittsburgh. Once aggregated at a week level, we have enough data that is not noisy to train our models.

Apart from Pittsburgh's historical 311 snow removal request records, we also gathered weather data from 2015-04-01 to January 2025 utilizing the free API information website called Open-meteo. The range we choose is based on and in accordance with the 311 data time range. The dataset contains 12 variables, and covers features such as maximum temperatures, minimum temperatures, average temperatures (all temperatures are in ℃), precipitation of snow (mm) and wind speed (m), etc.

To reflect differences in exposure and infrastructure burden, we also constructed a traffic intensity feature for each neighborhood. Using PennDOT sensor data from 2017 to 2021 which is only available through 2021, we first calculated the average daily car traffic by summing vehicle counts across valid days and dividing by the total number of days with available data. This average was then normalized by the geographic area of the neighborhood (in square miles), yielding a per-square-mile traffic density value. This traffic intensity measure provides a proxy for mobility-related demand, capturing the idea that areas with denser traffic may experience higher operational strain during snow events and potentially more service requests.

6. Details of solution

6.1 Feature Engineering

To support neighborhood-level snow complaint prediction, we constructed a structured panel dataset at the daily × neighborhood level. Each row represents a single neighborhood on a specific date. We engineered features based on four core categories as follows. All lagged and rolling features were computed within each neighborhood group to preserve temporal ordering and avoid data leakage.

Feature	Definition
date	Date of observation
neighborhood	Neighborhood name
daily_request_count	Target variable: number of snow-related 311 complaints
SNOW	Snowfall (cm)
PRCP	Precipitation (mm)
TMAX	Maximum temperature (C)
TMIN	Minimum temperature (C)
TAVG	Average temperature (C)
weekday	Day of the week
is_weekend	1 if Saturday/Sunday, else 0
is_weekday	1 if Monday-Friday, else 0
month	Month of the year
is_snow_season	1 if month is Nov-Mar, else 0
avg_daily_car_traffic_2017_2021	5-year average daily car traffic
sqmiles	Neighborhood area in square miles
lag_1d	Daily request count 1 day ago
lag_3d	Daily request count 3 days ago
rolling_mean_3d	Mean of daily requests over past 3 days
rolling_std_3d	Standard deviation of daily requests over past 3 days
lag_1y_same_week	Same ISO week average complaints from previous year

Table 1: list of all features in snow_removel_feature_engineered.csv

1. Temporal Features: We included calendar-based variables to capture seasonality and behavioral patterns:

• month: numeric month (1–12)

• is_weekday: 1 if Monday–Friday, 0 otherwise

• is_snow_season: 1 if the month falls between November and March

2. Weather Features: Daily weather variables were sourced from historical records and include:

• SNOW: snowfall accumulation (cm)

• PRCP: precipitation (mm)

• TMAX, TMIN, TAVG: maximum, minimum, and average temperatures (°C)

3. Lag and Rolling Features: To capture temporal dependencies and short-term dynamics in service demand, we created:

• lag_1d, lag_3d: complaint counts 1 and 3 days prior

• rolling_mean_3d, rolling_std_3d: average and volatility over the previous 3 days (excluding today)

• lag_1y_same_week: the average complaint volume from the same ISO week in the previous year

4. Spatial and Traffic Features: To capture demand heterogeneity across areas, we included static neighborhood attributes:

• avg_daily_car_traffic_2017_2021: five-year average of daily traffic volume per neighborhood

• sqmiles: physical area of the neighborhood (in square miles)

6.2 Model Selection and Training

We initially explored traditional time-series forecasting models such as SARIMA and Facebook Prophet due to their popularity in temporal prediction. But these models proved to be unsuitable for our task. Since both SARIMA and Prophet assume smooth seasonalities and continuous trends, they did not do well on our data because of the sparse, bursty nature of our 311 snow complaint data. There are days that have zero complaints while storm periods show big spikes.

Additionally, these models operate best at the aggregate (citywide) level and are not well-suited for high-resolution, multi-entity forecasting such as neighborhood-level daily predictions. From a computational standpoint, training separate SARIMA or Prophet models for each neighborhood was time-consuming and inefficient on local hardware, especially given the panel structure of our dataset. LightGBM allowed us to model complex interactions across a wide range of feature types which included temporal, weather, and traffic-related variables, making it well-suited for our task. It also supported fast training and tuning on local machines without the need for advanced hardware. LightGBM had the added advantage of interpretability via SHAP values which allowed us to understand which factors most influenced demand predictions.

Our final model is a LightGBM (LGBM) regressor, trained with features derived from historical request patterns and weather data, including rolling means and recent snowfall. We conducted hyperparameter tuning using randomized search with 5-fold time-series cross-validation, optimizing for MAE. Key parameters such as num_leaves, max_depth, and learning_rate were adjusted to balance accuracy and overfitting risk. We evaluated model performance using three key metrics. Model accuracy was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). Each of these metrics offered a slightly different perspective on model performance and had its own strengths and limitations depending on the application context.

6.3 Model Validation

To evaluate model performance, we adopted a time-based hold-out validation strategy aligned with real-world deployment. We trained our models on historical data from 2015 to 2022 and reserved the years 2023 and 2024 (extending into early 2025) as the test set. This approach preserved the temporal sequence of events and ensured that future data was never used in training, mimicking operational forecasting conditions.

We benchmarked our model against three intuitive, time-based baselines:

- Lag-1: uses the previous day's complaint volume
- Lag-7: uses the same day of the previous week
- Year-over-Year (YoY): uses the same ISO week from the previous year

While these baselines are simple and interpretable, they lacked the flexibility to account for neighborhood-level dynamics or respond to sudden changes in weather. For example, Lag-1 had an MAE of approximately 30, meaning it deviated from actual complaint volumes by 30 requests per day on average—providing a useful but limited benchmark.

We evaluated model performance using three key metrics:

- MAE: captures the average prediction error in absolute units; intuitive and robust to outliers.
- RMSE: penalizes large errors more heavily, useful for highlighting underperformance on high-demand days.
- MAPE: expresses error as a percentage of actual values but becomes unstable when true values approach zero—a common issue in low-complaint neighborhoods or dry weather periods.

Our final LightGBM model significantly outperformed all baseline models in both MAE and RMSE, particularly during high-demand storm periods. This demonstrated its reliability for both routine operations and rare, high-impact events. Lastly, this validation framework not only confirmed the predictive strength of our approach but also mirrored how the model would be deployed operationally to inform daily staffing, salt distribution, and routing decisions.

The complete data processing and modeling pipeline—including data cleaning, feature generation, model training, and evaluation—is documented and available in our GitHub repository(https://github.com/moidulhn/pittsburgh-snow-removal-forecasting).

7. Evaluation and Comparison

7.1 Simple baseline method

For starters, we try the simple time-based models to examine the result as well as explore the dataset.

	Model	MAE	RMSE	MAPE (%)
0	Lag-1	30.61875	62.887747	216.525708
1	Lag-7	50.35000	98.273280	997.778886
2	YoY	65.96875	123.355356	2166.380477

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Table 2: MAE, RMSE and MAPE result of Lag-1, Lag-7 and YoY simple time-based baseline

As seen from the table above, among simple time-based baselines, Lag-1 provides the most stable and useful benchmark. It is in accordance with our intuitions that weather is more closely related with the weather the day before compared with the average weather in the past 7 days and the past year. This suggests that yesterday's request count is still somewhat predictive, though not highly accurate. However, Lag-7 and YoY perform very poorly, especially in terms of MAPE. Week-to-week patterns may be unstable or irregular, especially in snow seasons. Year-over-year patterns are not reliable, likely due to shifting weather or demand trends. The high MAPE values (especially > 2000%) mean that many actual values are small (e.g. close to 0), which inflates percentage errors. The result motivates us to use machine learning models like LGBM for improved accuracy.

7.2 Adopting LightGBM method: citywide results

After we ran the LightGBM model, we plot the predicted vs. actual results graphs, comparing the results using model LightGBM with Lag-1 baseline, Lag-7 baseline and YoY baseline. We also made comparisons from the following four dimensions: Citywide daily requests, the results are as follow:

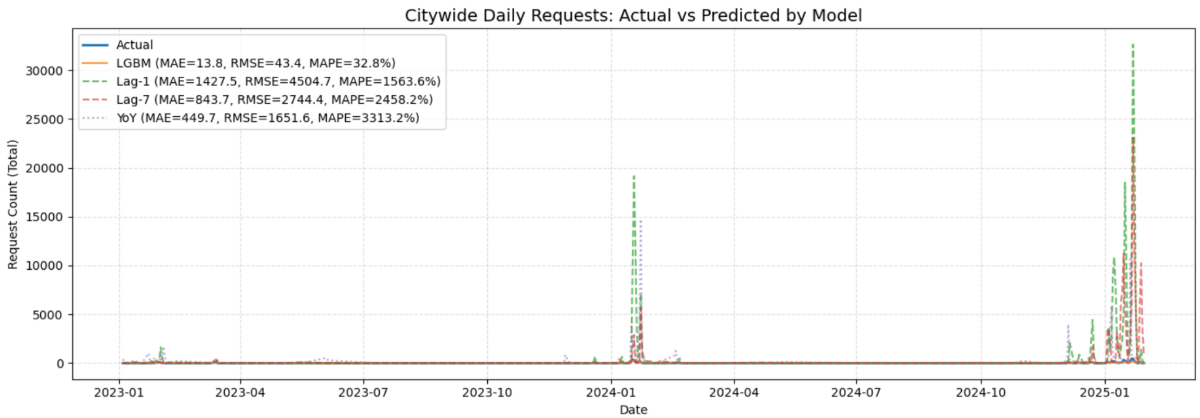

Figure 1: a line chart comparing different models in citywide daily requests prediction

The citywide plots show that the LGBM model closely follows the actual request trends across the entire testing period, including sharp peaks during major snow events in January 2024 and January 2025. In contrast, the baseline methods, particularly Lag-7 and YoY, frequently overestimate the request volume, especially during extreme weather conditions. This suggests that simple temporal baselines struggle to adapt to sudden shifts in demand. LGBM, nevertheless, maintains a stable response without extreme spikes, indicating better generalization and resistance to overreacting to past extremes.

It is not difficult to understand that for most of the days in Pittsburgh there's no snow, and therefore leads to no or little requests about snow removal. Hence, the majority of the data are around or at zero. However, we do notice three peaks in the plot above, and we are also curious about the models' performance in days with tremendous results. Thus, we extract the dates, and zoom in to see the particular predicting results in these three periods of time.

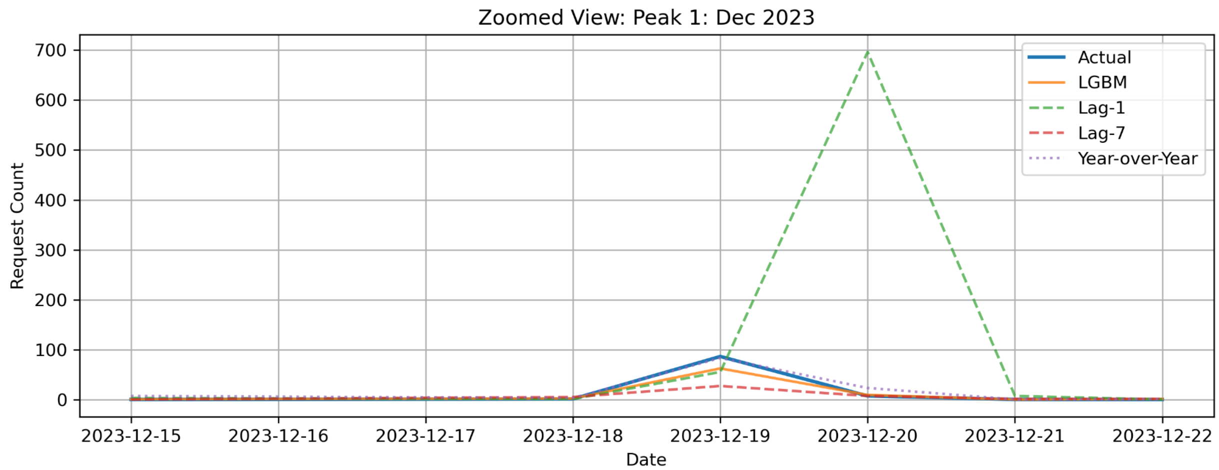

Figure 2: a line chart comparing different models in citywide daily requests predictions in Dec 2023

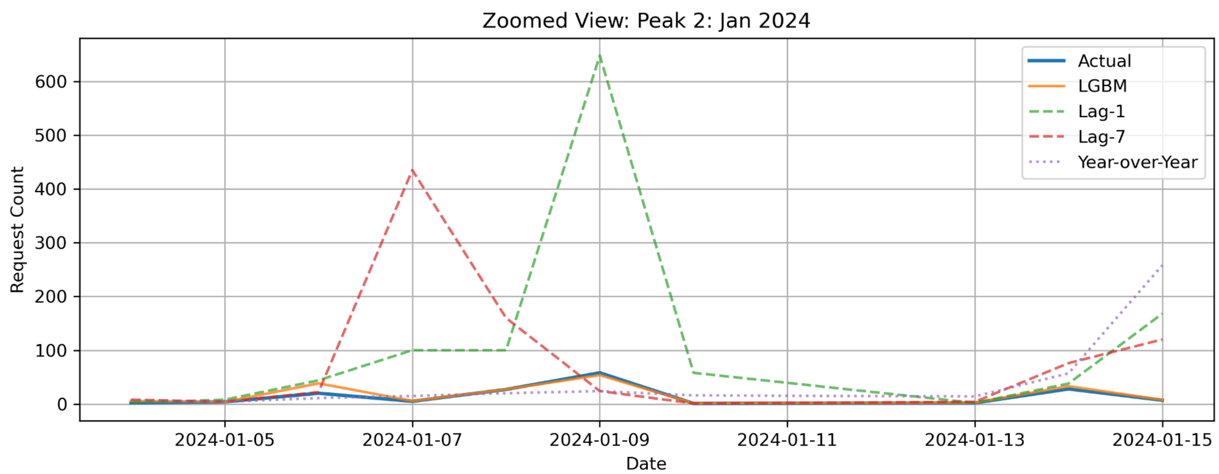

Figure 3: a line chart comparing different models in citywide daily requests predictions in Jan 2024

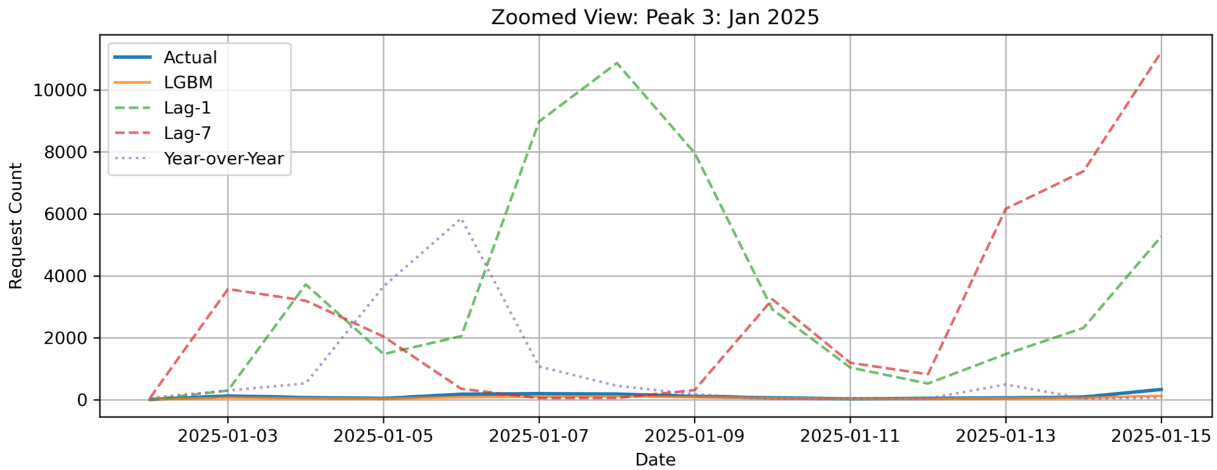

Figure 4: a line chart comparing different models in citywide daily requests predictions in Jan 2025

Zoomed-in views of three peak windows (December 2023, January 2024, and January 2025) further highlight these differences. The LGBM model consistently tracks the real-world request spikes with smoother and more calibrated responses. Lag-based baselines, especially Lag-1 and Lag-7, often react too strongly to previous high values, leading to exaggerated predictions. The YoY baseline performs worst, failing to reflect real-time variation and lagging far behind current patterns. These results emphasize that recent weather and request history—not historical averages from previous years—are more reliable predictors of current demand.

7.3 Adopting LightGBM method: neighborhood results

Neighborhoods vary widely in population, geography, and request volume. MAE provides a unit-consistent and intuitive measure of average prediction error. By looking at AME, we can evaluate our prediction like "On average, we were off by 10 requests in this neighborhood." Unlike RMSE, MAE doesn’t disproportionately penalize large errors, which might be misleading in low-volume neighborhoods where even one spike could distort RMSE. MAPE is also not reliable in this case in that many neighborhoods have days with zero or near-zero requests, which makes MAPE unstable or misleading.

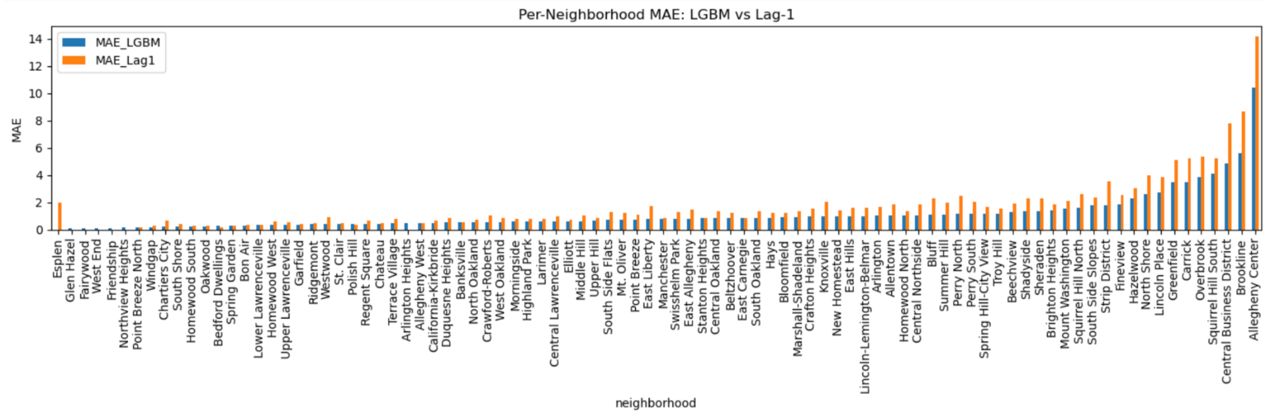

Figure 5:a bar chart comparing different models in neighborhood overall requests prediction

As we can see from the graph above, at the neighborhood level, the LGBM model demonstrates strong predictive power across diverse regions. A per-neighborhood MAE bar chart shows that LGBM outperforms Lag-1 in nearly all areas, with particularly large improvements in high-demand neighborhoods or with broader ranges such as Allegheny Center, Downtown, and Brookline. In contrast, baseline errors remain high in these neighborhoods, underscoring their inability to generalize well in more complex, variable environments. This suggests that LGBM is especially effective where accurate forecasting is most operationally critical.

7.4: Adopting LightGBM method: seasonal trend results (monthly and weekly)

Seasonal trends are about capturing patterns across months and weeks, and MAE gives a clear sense of whether predictions were generally higher or lower than actuals on average per month. It allows us to compare forecast accuracy across months without worrying about scale differences or the exaggeration of large spikes. RMSE, however, could be skewed by storm weeks in one or two months, making inter-month comparison less fair. MAPE has the same problem with what we state in the neighborhood section. Again, it is unreliable in low-volume summer months when actual values drop close to zero.

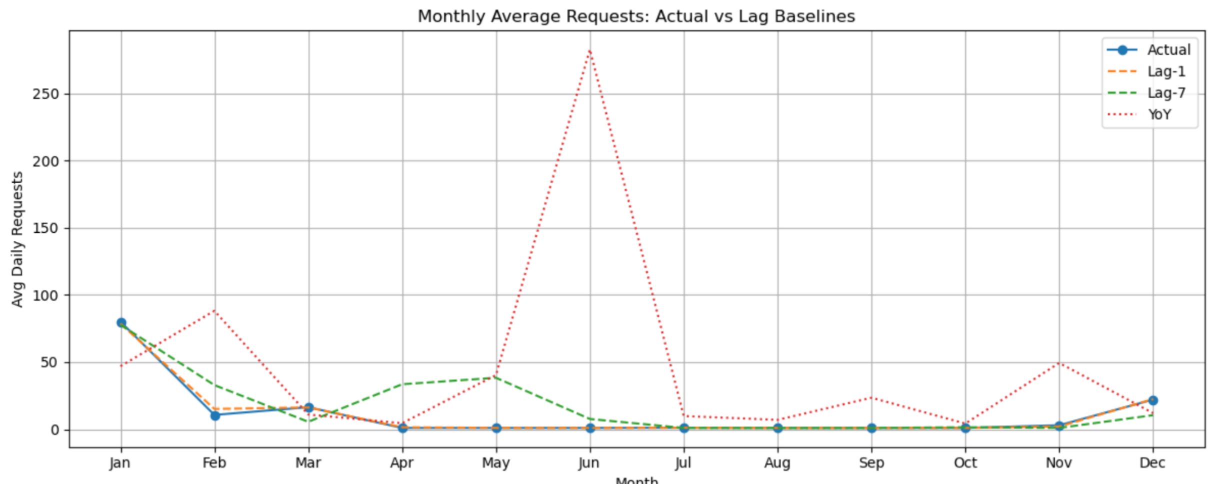

Figure 6: a line chart of monthly average predictions analysis

The monthly average request plot confirms that snow removal demand is highly seasonal, with activity peaking in January and February and falling close to zero in warmer months. The Lag-1 baseline captures this broad pattern reasonably well, but Lag-7 and YoY diverge significantly from actual trends. YoY in particular shows erratic spikes, including a severe overestimation in June. These findings highlight the limitations of baselines that rely on rigid temporal lags or historical analogs. By contrast, LGBM leverages short-term signals to stay responsive to real-time shifts in weather and behavior.

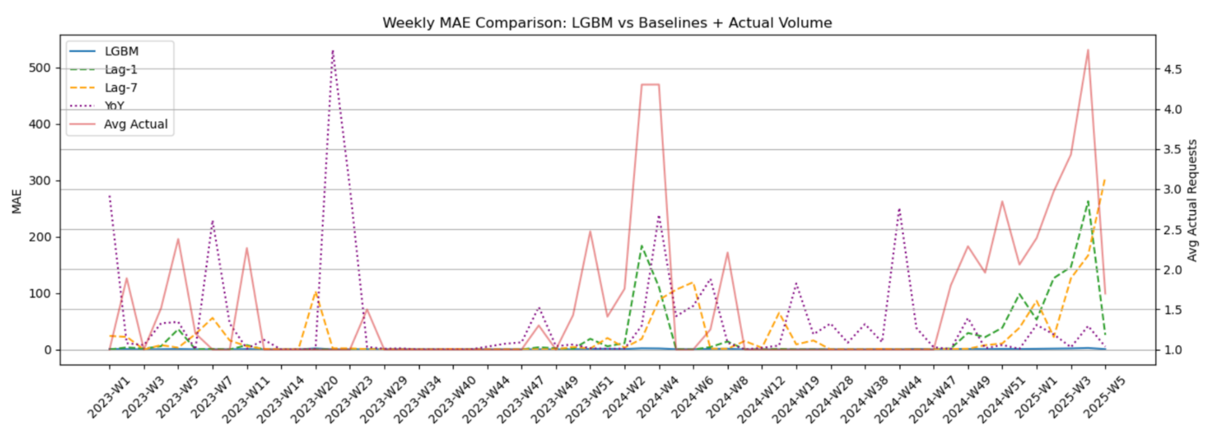

Figure 7: a line chart of weekly average predictions analysis

The weekly MAE analysis reinforces this insight. LGBM maintains the lowest weekly prediction error throughout the test period, including during high-stakes snow weeks in early 2024 and 2025. Lag-1 performs adequately during normal periods but becomes unreliable during high-volume weeks. Lag-7 and YoY again exhibit volatile error patterns, with MAE exceeding 300 during storm periods. This variability demonstrates that LGBM provides more consistent and dependable performance in operational forecasting scenarios where stability is essential.

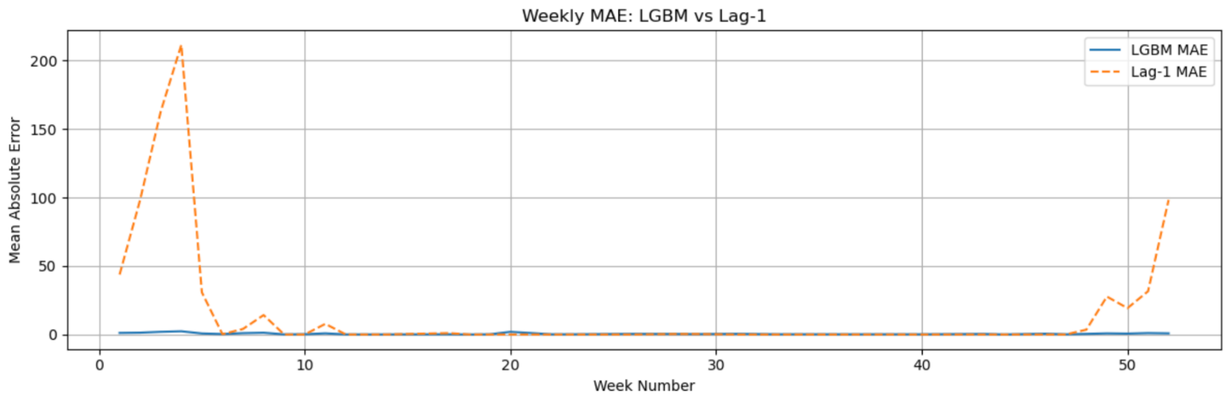

Figure 8: a line chart of weekly average predictions analysis of LGBM and Lag-1

As the previous graph is a bit chaotic, partially due to the inaccurate predictions of simple time-based baselines, we also extract only Lag-1 and LGBM models to compare their performances. Still, LightGBM outperformed Lag-1, especially in days with large requests of snow removal.

7.5: SHAP analysis of feature importance

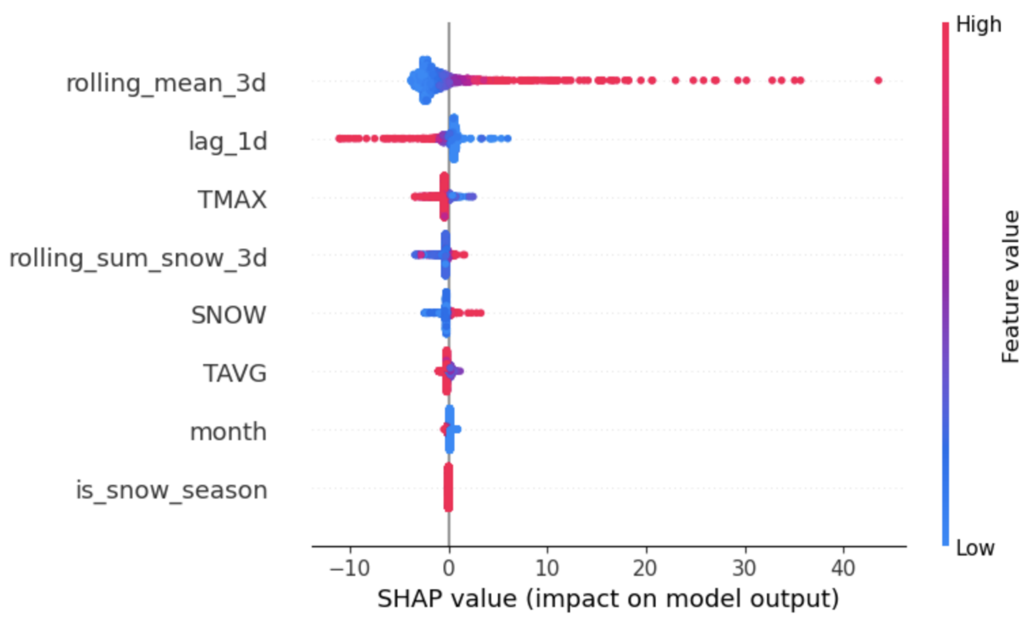

Figure 9: SHAP analysis of feature importance

Finally, SHAP analysis of feature importance offers valuable insight into the model's decision process. The most influential features include short-term historical patterns (rolling_mean_3d and lag_1d) and weather variables such as TMAX, TAVG, and SNOW. Calendar features like month and is_snow_season are relatively less important. This reinforces the conclusion that recent snow activity and short-term trends are far more informative than coarse seasonal indicators. Importantly, the SHAP values confirm that LGBM not only performs well empirically but also aligns with domain knowledge about how snow events drive public service demand.

8. Discussion

Our approach demonstrates that an ML model can predict snow-removal requests for the city of Pittsburgh at a daily frequency for each area. This can help DPW meaningfully improve the city's ability to anticipate snow removal requests for each neighborhood. The model's strong performance, especially during peak snow periods, suggests utility in DPW's operations by shifting from a reactive to a proactive approach, ultimately leading to an increase in efficiency, equity and responsiveness.

The error metrics indicate a robust model which consistently predicts daily request counts with reasonable accuracy, providing significant improvements over baseline methods such as Lags and YoY averages. Feature importance analysis with SHAP reveals that complaint history and snowfall levels are the most important predictors of future demand. Although this seems obvious, operational planning is still difficult due to how different areas are affected by the same amount of snowfall. Additionally, the model was able to closely follow larger demand surges during heavy snowfall without overestimating.

The results have great utility for DPW. Daily, neighborhood-level forecasts can inform decisions about where to deploy plows and salt trucks, how to schedule crew shifts, and when to initiate communications with residents to warn them. In practice, this would help DPW preemptively address likely problem areas, reduce backlogs, and minimize service inequities during storms.

9. Policy Recommendations

Following the successful development and validation of our predictive model, we recommend that the City of Pittsburgh shift from experimentation to operational use. The model has demonstrated strong accuracy in forecasting neighborhood-level complaint volumes several

days in advance, outperforming traditional baselines and capturing both seasonal and short-term variation in snow-related demand.
To translate these predictive insights into practical improvements in city operations, we propose a set of policy recommendations focused on integration, equity, communication, and strategic planning.

Integrate forecasts into daily operations: Model outputs should be systematically incorporated into pre-storm planning workflows coordinated by the Department of Public Works (DPW) and Department of Mobility and Infrastructure (DOMI). The predictions provide a quantitative basis for optimizing operational variables such as plow crew shifts, equipment staging, and salt pre-deployment. High-demand days, as flagged by the model's daily output, can trigger early activation thresholds or route adjustments before complaints begin to accumulate.

Use forecasts to track service equity: Because the model generates geographically disaggregated forecasts, it can be used not only for deployment planning but also for retrospective fairness audits. Comparing predicted service needs with actual plow activity or response volume allows the city to identify persistent mismatches, particularly in historically underserved neighborhoods, and adapt future policies or dispatch logic accordingly.

Enable proactive communications: Forecast results can be shared with residents through city dashboards or service alerts, helping to set expectations, communicate timelines, and reduce duplicate 311 reports during storm peaks. Transparent sharing of "forecasted demand zones" may also reduce frustration among residents.

Embed into long-term infrastructure planning: Aggregated model outputs across seasons should be used to inform budget planning, procurement of equipment, and contractor agreements. Areas with historically high predicted and actual demand may warrant route redesign or additional staffing.

Begin phased deployment with pilot districts: We recommend that the model be tested operationally in 1–2 snow-sensitive council districts during the next winter season. This pilot phase will allow departments to refine data workflows, train staff, and assess implementation challenges before expanding the system citywide.

By leveraging this predictive framework, the City of Pittsburgh can move toward a more proactive, equitable, and data-informed model of winter service delivery.

10. Limitations, caveats and future work

This project builds on a robust dataset of historical 311 complaints, combined with weather, temporal, and spatial features. However, several limitations remain that affect both accuracy and real-world applicability.

First, the dataset contains temporal and geographic imbalance. While it includes more than 32,000 snow-related requests across nine winters, certain storm events generated extreme complaint spikes that may bias the model. In addition, complaints are not uniformly distributed across neighborhoods, which limits the ability to evaluate fairness across all areas.

Second, the model estimates public demand but does not capture actual service delivery. Without access to plow activity, salt usage, or response timestamps, we cannot assess how well the model aligns with operational outcomes. Integrating data from city fleet GPS or service logs would enable stronger validation and accountability.

Third, Pittsburgh's complex topography and neighborhood-level microclimates introduce variability that is difficult to capture with aggregated features. Although weather variables are included at the city level, they may not reflect conditions on individual streets. Future iterations could incorporate higher-resolution environmental data or real-time street-level sensors.

Finally, the model has not yet been tested in a live deployment context. While performance was validated using recent holdout years, its integration into real-time decision-making workflows has not been evaluated. Future work should explore operational testing with city departments, adaptive model retraining, and user-facing forecast tools to support staff decisions.

Despite these limitations, the forecasting system offers a scalable and interpretable framework for improving snow response planning and service equity in Pittsburgh. Continued refinement will enhance its readiness for operational use.